



GS4400

RS-485 Transmitter Communications Guide



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Intended Use

The pressure transmitter is designed for monitoring liquid and gaseous media within the limits specified in these instructions.

For this purpose, the pressure transmitter converts pressure applied to the pressure connection into an electrical signal.

Please pay attention to the correct method of the sealing and process connection for the transmitter (See electrical connections and threads)

Please take into account the environmental conditions such as temperature, humidity and atmospheric pressure, etc.

Should be handled by qualified technical personnel only.

Limitation of Liability

The manufacturer is not liable for damage resulting from non-observance of these operating instructions.

Safety Symbols

The explanations of the symbols used in these operating instructions are listed below;

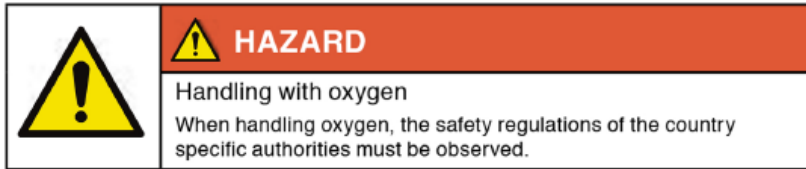
	 HAZARD
	Type and source of the hazard! Consequence - Hazard potentially resulting in death, personal injuries and damage to property
	 WARNING
	Type and source of the hazard! Consequence - Hazard potentially resulting in minor personal injuries and damage to property
	 CAUTION
	Type and source of the hazard! Consequence - Hazard potentially resulting in damage to property
	 NOTICE
	Type and source of the hazard! Notes, information and recommendations e. g. for trouble-free operation

Safety

In addition to the safety instructions included in this operation manual, local safety regulations apply.

Use the product in its original state only, without making any unauthorised changes.

Prior to the installation of the product all transport materials, such as protective covers, caps or cardboard must be removed.

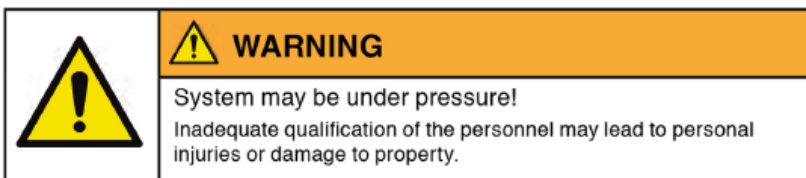


Selection of personnel

Only authorised and qualified personnel who have experience and knowledge of the country-specific regulations, as well as the applicable standards and directives, may be used for installation and operation.

Installation and commissioning must be carried out in accordance with these operating instructions.

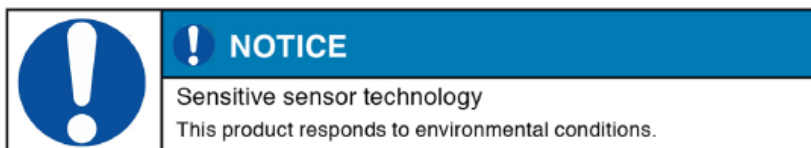
The pressure transmitter is regarded as “electrical equipment”



Product Application

The pressure transmitter is intended to monitor liquid and gaseous media. For proper and safe application of the product. It is important to follow the specifications and warnings described in these operating instructions.

In addition to this operating manual, the specific safety regulations in the country in which the product is used, apply.



Please comply with the specified limit values such as pressure ranges, overpressure safety, operating voltage and temperatures

Example of limit values

Overpressure safety: 4x for 0.5 bar range; 2x for ranges -1 bar to 600 bar; 1.5x for 1,000 bar range; 1.1x for 1,500 bar range; Maximum pressure change rate: <1,0 bar/ms

- Maximum pressure change rate <1,0 bar/ms
- Please take into account the prevailing environmental conditions (temperature, humidity, atmospheric pressure, etc)
- Please use the product in its original state only. Do not carry out any unauthorised modifications.
- Remove any transport materials such as protective covers, caps or cardboard.
- Material compatibility should be considered. If in doubt, contact ESI sales.

Maintenance and Care

This pressure transmitter is maintenance-free. Nevertheless you should check the pressure transmitter regularly for excessive dirt and defects.

Recalibrations cycle depends on own industry guidelines.

	 WARNING
	System may be under pressure! Loose parts and escaping media may cause personal injuries to the personnel present. Ensure the pressure system is depressurised prior to starting any cleaning work.

Cleaning

Clean the device using a damp cloth.

- Never use sharp or hard objects to clean the device.
- Do not insert or push any objects into the pressure connection hole

	 NOTICE
	Repair work on the product must be carried out by the manufacturer only

Scope of Delivery and Storage

	 WARNING
	Damage to the product! Check the pressure transmitter for damage when unpacking. If the pressure transmitter is damaged, contact the manufacturer or your local dealer immediately.

Storage

	 WARNING
	Improper storage! Improper storage of the pressure transmitter may result in damage or malfunction of the product.

Ambient conditions during storage

- Temperature: +5 °C to +40 °C
- Store in dry conditions.

Malfunctions

In case of malfunctions, first check that the pressure transmitter is mounted correctly. The table shows the most common faults and malfunctions and how they can be rectified.

Error/fault	Possible cause	Possible remedial action
No output signal	- no supply voltage - line interrupted/disconnected - polarity reversal	- switch on supply voltage - check cable - adjust/reset connection
Zero point signal outside of specification	- operating outside of temperature range - measuring cell damaged - pressure connection hole blocked	- adhere to temperature range - replace transmitter - clean bore hole
Signal span outside of specification	- operating outside of temperature range - measuring cell damaged, e. g. by overpressure - pressure connection hole blocked	- adhere to temperature range - replace transmitter - clean bore hole
Output signal fluctuates / noises	EMC interference sources outside the specification	- eliminate or reduce EMC interference sources - additional EMC precautions (e. g. shielding)
Output signal does not change	- measuring cell damaged, e. g. by overpressure - Electronics failure, e. g. by short circuit	replace transmitter

Recycling

Dispose of the product and transport packaging materials according to applicable disposal regulations specific to the respective country. Upon request, used parts can be taken back free of charge and disassembled, disposed of and recycled in accordance with the legal regulations.



	 WARNING
	After removal! After removal, there may be residual media in and at the pressure connection that can endanger persons and the environment. Please take appropriate precautions.

General Information

These operating instructions describe the safe and efficient handling and operation of the pressure transmitter or transducer, and must be carefully read and understood prior to use.

Compliance with the specifications regarding safety and operation contained in these instructions is a prerequisite for safe operation.

These operating instructions are part of the product. Failure to comply with the operating instructions will void the manufacturer's liability in the event of damage.

For future reference, keep these instructions in an accessible location.

Product Coding Protocol

The standard ESI Technology Ltd coding for pressure transducers and transmitters is as follows:

MODEL	VARIANT	PRESSURE RANGE	PROCESS CONNECTION

Coding detail

Model Code: **GS4400** General Purpose Pressure Transmitter with RS-485 Interface

Variant Code: **B** M12 Connector

Range Code:

V001	0-1 bar Vac
0006	0-6 bar
0025	0-25 bar
1500	0-1500 bar

Process Connection:

AB	¼" BSP male
AC	½" BSP male
AM	¼" NPT Male
AN	½" NPT Male

Order Number Example: **GS4400B1500AB**

ESI Product Instructions - General

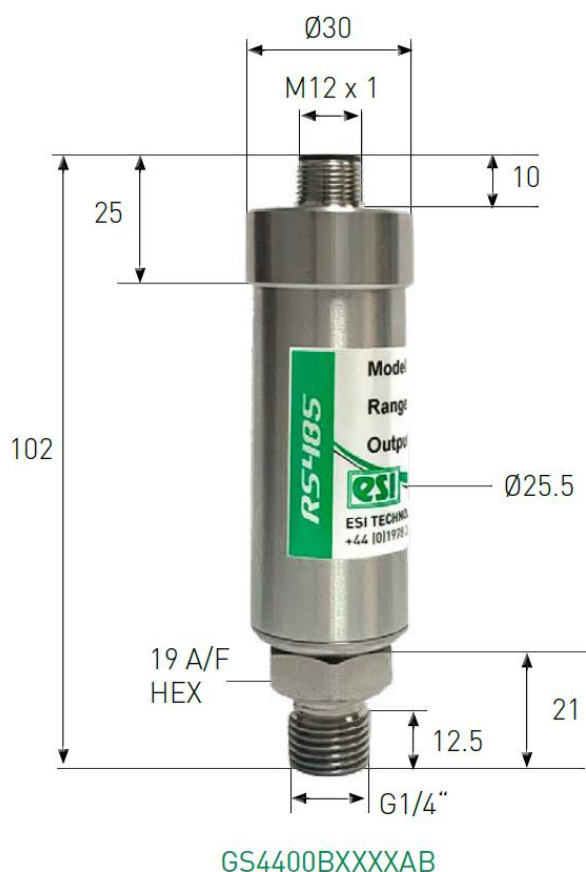
INTENDED USE	
	Please refer to datasheet, calibration certificate and read installation instructions before starting installation.
	The device must be used within the specified pressure range, temperature and supply voltage.
INSTALLATION	
	During installation please comply with the relevant national guidelines.
	Installer must be technically competent and familiar with pressure monitoring technology.
MECHANICAL INSTALLATION	
	Ensure system is not pressurised before installation.
	Ensure that the measurement cell is not damaged during installation.
	Use either a correct sized AF wrench on the hexagon with tightening torque in accordance with the table. The customer must ensure that the pressure seal is suitable for the application.
	Tightening torque setting guideline in table overleaf.
ELECTRICAL INSTALLATION	
	Make mechanical installations first so as not to twist the cable.
	Refer to product calibration certificate for wiring diagram.
	Ensure electrical connector and cable gland are securely fitted and sealed.
MAINTENANCE	
	Recalibration interval depends on own industry guidelines.
	Do not insert pointed or hard objects in to the pressure port.
	Repairs must be performed by manufacturer only.
REMOVAL / END OF LIFE	
	Ensure system is depressurised before removal.
	Remaining media in pressure port may be hazardous.
	WEEE can be returned to factory at end of life for recycling.

GS4400 SERIES

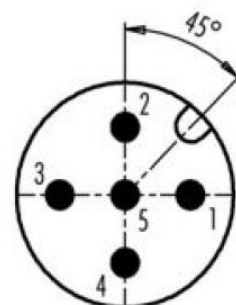
Disclaimer

ESI Technology Ltd operates a policy of continuous improvement and product development. We reserve the right to change specification and operating instructions without prior notice.

Please observe applicable safety regulations when installing or removing the pressure transmitter.

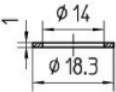
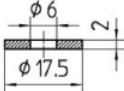
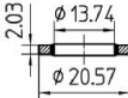
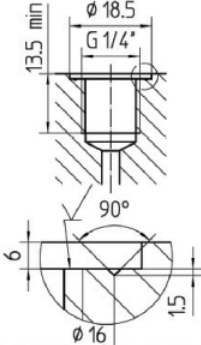
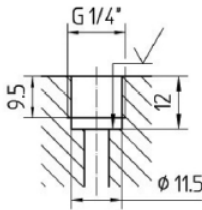
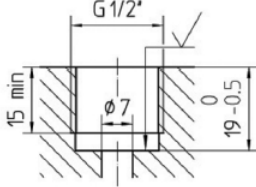
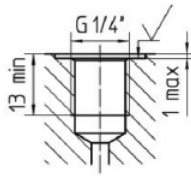


Pin No	Designation
1	RS485(B)
2	RS485(A)
3	Common Ground
4	DS Power IN
5	Analogue Output
Case	Case GND



Connections and Tightening Torques

GS4400 SERIES

MODEL NUMBER				
Process Connection	G1/4" (BSPP)		G-1/2 EN 837-1	G1/4 (BSPP)
Sealing Method	Copper Washer (annealed)		Copper Washer (annealed)	Bonded Seal ("Dowty") < 1000 bar
Counter Part	Part No.: 20068  Part No.: 9727 		Part No.: 20942 	
	 		 	

Tightening Torques

* The torque values apply only to test conditions and are a guide only. Tightening torques for installation depend on many factors, including materials, lubrication, coating and surface treatment. If in doubt the manufacturer is to be consulted.

** NPT threads need an additional sealant to BA EN 751 (e.g. PTFE tape)

*** Tightening torque and maximum pressure should be in accordance with high pressure pipe supplier's documentation.

Materials of the mating part should be selected to suit pressure and media requirements.



Thread Type	Torque Settings *				
	P ≤ 100bar	100bar < P ≤ 400bar	600bar < P ≤ 1500bar	2000bar	4000bar/ 5000bar***
G1/4" (bSPP), 1/4-18 NPT**	Max. 35 Nm			---	---
G1/2" (BSPP), 1/2-14 NPT**	Max. 50 Nm				
M16 x 1.5	---	---	30-35 Nm	32-35 Nm***	

Introduction

This guide describes the RS-485 interface and protocol used by the ESI RS-485 series of pressure transmitters.

The transmitters use a standard 2-wire RS-485 interface with a proprietary ESI protocol. This protocol and the basic principles of RS-485 communication are outlined in this document.

RS-485 Principles

This section covers the basic principles of the RS-485 communication bus in order to properly communicate with the connected transmitters.

1. Physical

The RS-485 connection is a hardware design that allows multiple transmitters to be connected to the same bus, reducing cabling and increasing transmission distance.

The RS-485 bus uses a twisted pair of wires and a differential digital signal to communicate between the master and the devices. The bus can be configured in multiple topologies and under ideal conditions can allow transmission up to 1200m.

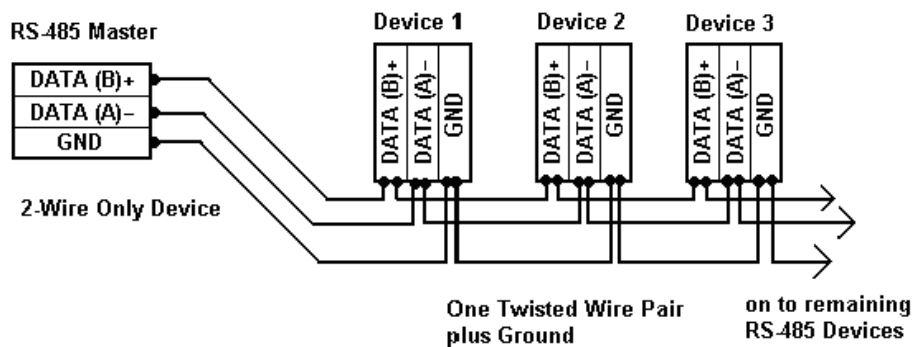


Fig. 1
2-Wire RS-485 Connections

The ESI RS-485 transmitters use a 2-wire or half-duplex arrangement for communication, reducing cabling requirements and connector cost but does not allow bidirectional communication. This is often easier to implement for users creating custom software interfaces.

2. Master-Slave Principal

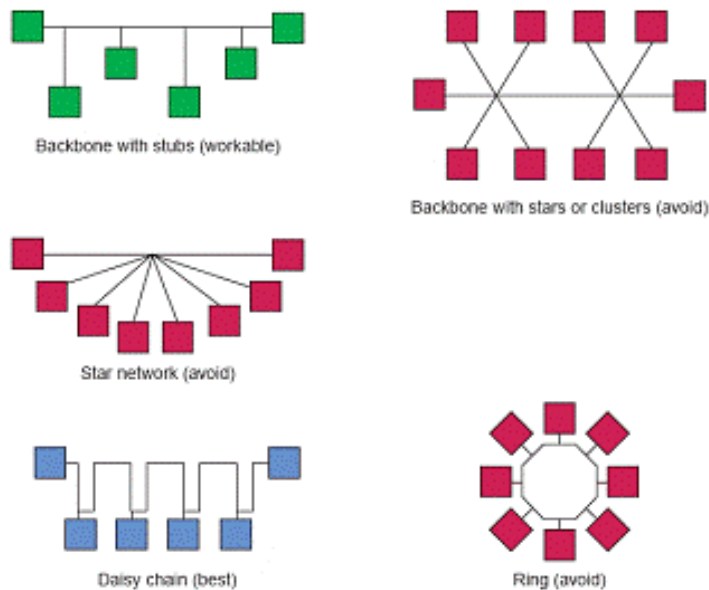
The transmission over RS-485 depends upon a master-slave principle, in which there is only ever one master (the controller) and one or more slaves (the devices/transmitters).

The master controls the data exchange by sending out instructions or requests to the connected devices. The slave devices only have a response function and will not transmit a message unless first instructed to by the master controller.

This prevents a condition known as bus contention where multiple transmitters attempt to place data on the bus, resulting in lost data at best and damage to the devices at worst.

3. BUS Topology

The bus topology or layout, is the configuration of the RS-485 bus and the connected devices. Multiple bus topologies exist, for the best operation a daisy chain topology is recommended.



Depending on the specifics of the implemented bus and the other devices connected, it may be necessary to connect a termination resistor to the end of the bus, between the A and B wires. The termination resistor prevents signal reflection. As a general rule, the value of the termination resistor should match the characteristic impedance of the twisted pair cable. Most often a 120Ω termination resistor is appropriate.

4. Transmission Mode

The transmission mode uses an ASCII based system for easy readability when using a terminal.

The communication data settings are listed below:

Baud Rate (Default):	57600
Data Word:	8-bit
Parity Bit:	None
Stop bit:	1

The baud rate may be changed but will require the master controller to be changed as well, such that both are matched.

The available baud rates are: 9600, 14400, 19200, 28800, 38400, 57600, 115200, 230400.

5. Transmitter Addresses

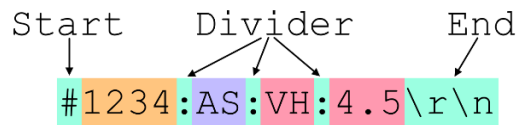
The communications protocol uses a non-standard addressing method, in which the transmitters serial number is used to differentiate incoming messages. Each message from the master must contain the correct serial number otherwise the transmitter will ignore it.

There are no broadcast commands and all correctly addressed messages will be met with a response from the transmitter.

There is also one command used to identify connected transmitters, however it should only be used when just a single transmitter is connected to the bus.

6. Data Arrangement

All outgoing (master to slave) data is arranged using a start character, an end character and dividers between each section.



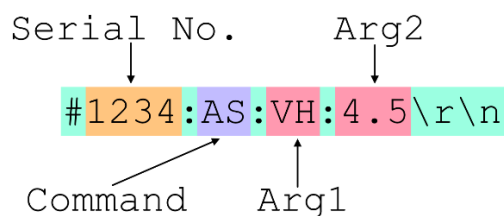
The start character is a pound sign (#).

The dividers are colons (:).

The end of line character can be either carriage return, line feed or both (\r\n).

If for example an argument is not used, the associated divider is not required, e.g. a command with no arguments will only contain one divider.

All messages contain a serial number, a command and up to 2 data or argument sections.



The argument sections are not always required but this depends on each command and will be explained in each command section.

All strings are a case sensitive. The serial number must not contain letters.

Messages returned from the device (slave to master) have the same structure, however they use a dollar sign (\$) as a start character instead of the pound sign.

\$1234:AS:VH:4.5\r\n

7. Error Treatment

If a message with an invalid address or command is sent then the transmitter will ignore it. If an invalid argument is sent then the transmitter will respond with the default or existing data in relation to the request.

Communication Commands

This section describes the commands available to interface with and control the ESI RS-485 transmitters.

In the examples given, the EOL characters (\r\n) are not shown. Messages sent from the master controller to the device are coloured red and include the # start character, whereas messages returned from the devices are coloured blue and include the \$ start character.

Comments about the example are in green and follow “//” and would not be sent to the transmitter.

1. Identify Transmitter - ID

The ID command allows a single transmitter on a bus to identify itself with its own serial number. Even if the transmitters serial number is not known, the ID command can be used to request the sensor respond, thus identifying its self.

The ID command is special for this reason and is the only command that can use 0 as the serial number to communicate with a device.

This should only be done if there is just one single transmitter on the bus, to prevent damage through bus contention.

If the command is addresses specifically to a connected sensor then only that sensor will respond, allowing the user to “ping” the sensor.

Command: ID

Args: <None>

Example: #0:ID //request ID from unknown transmitter
\$1234:ID

#1234:ID //ping a known transmitter
\$1234:ID

2. Transmitter Info - TI

The TI commands returns some human readable information about the transmitter, such as the firmware revision, pressure range, baud rate and manufacturing/calibration dates.

Command: TI

Args: <None>

Example: #1234:TI //request transmitter information
\$1234:Serial Number - 30719
\$1234:Firmware Rev. - 1.00
\$1234:Pressure Range - 0.00 bar to 2000.00 bar
\$1234:Baud Rate - 57600
\$1234:Man. Date (Y/M/D) - 2021/04/20
\$1234:Cal. Date (Y/M/D) - 2021/05/07

Each line includes an EOL character.

3. Sample Pressure - SP

This command returns raw pressure sample data from the transmitter's ADC, which can be processed separately if required. The first argument can be set to request the data in decimal or hexadecimal format, or left blank to return the data in decimal format by default.

The data returned will be an unsigned integer between 0 and 16777215 (represented as 0xFFFFF in hexadecimal).

Command: SP

Args: <None>, HEX, DEC

Example: `#1234:SP //request pressure sample data`
`$1234:SP:8393722`

`#1234:SP:DEC //request data (decimal)`
`$1234:SP:DEC:8393722`

`#1234:SP:HEX //request data (hexadecimal)`
`$1234:SP:HEX:0x8013FA`

4. Sample Temperature - ST

This command returns raw temperature sample data from the transmitter's ADC, which can be processed separately if required. The first argument can be set to request the data in decimal or hexadecimal format, or left blank to return the data in decimal format by default.

The data returned will be an unsigned integer between 0 and 16777215 (represented as 0xFFFFF in hexadecimal).

Command: ST

Arg1: <None>, HEX, DEC

Example: `#1234:ST //request pressure sample data`
`$1234:ST:13600564`

`#1234:ST:DEC //request data (decimal)`
`$1234:ST:DEC:13600564`

`#1234:ST:HEX //request data (hexadecimal)`
`$1234:ST:HEX:0xCF8734`

5. Read Pressure - RP

The RP command returns the calculated pressure from the transmitter in a choice of units. The unit is chosen by setting the first argument. The units are case sensitive and must match exactly.

The data returned is in the format of a floating-point value.

Command: RP

Args: <Pressure Units>

```
Example: #1234:RP:Bar      //request pressure reading in Bar
         $1234:RP:Bar:1.562

         #1234:RP:psi      //request reading in psi
         $1234:RP:psi:22.654895

         #1234:RP:Pa       //request reading in Pascals
         $1234:RP:Pa:156200
```

6. Read Temperature - RT

The RT command returns the calculated temperature from the transmitter in a choice of units. The unit is chosen by setting the first argument. The units are case sensitive and must match exactly.

The data returned is in the format of a floating-point value.

Command: RT

Args: <Temperature Units>

```
Example: #1234:RT:C        //request temperature in Celsius
         $1234:RT:C:18.145508

         #1234:RT:F        //request temperature in Fahrenheit
         $1234:RT:F:64.661914

         #1234:RT:K        //request temperature in Kelvin
         $1234:RT:K:291.295508
```

7. User Pressure Offset - PO

This command allows a user pressure offset to be applied to the calibrated pressure output. It can be used to effectively tare the transmitter pressure. If the message is sent without arguments then the transmitter will return its current user pressure offset. For the PO command, all pressure units are in Bar and the outgoing messages do not require units.

Command: PO

Args: <None>, <Pressure Offset in Bar>

```
Example: #1234:PO          //view current pressure offset
         $1234:PO:Bar:0.000000

         #1234:PO:15.12    //apply new pressure offset (Bar)
         $1234:PO:Bar:15.120000

         #1234:PO          //confirm new pressure offset
         $1234:PO:Bar:15.120000
```

8. User Temperature Offset - TO

This command allows a user temperature offset to be applied to the calibrated temperature output. It can be used to effectively tare the transmitter temperature. If the message is sent without arguments then the transmitter will return its current user temperature offset. For the TO command, all temperature units are in C and the outgoing messages do not require units.

Command: TO

Args: <None>, <Temperature Offset in C>

Example: `#1234:TO //view current temperature offset`
`$1234:TO:C:0.000000`

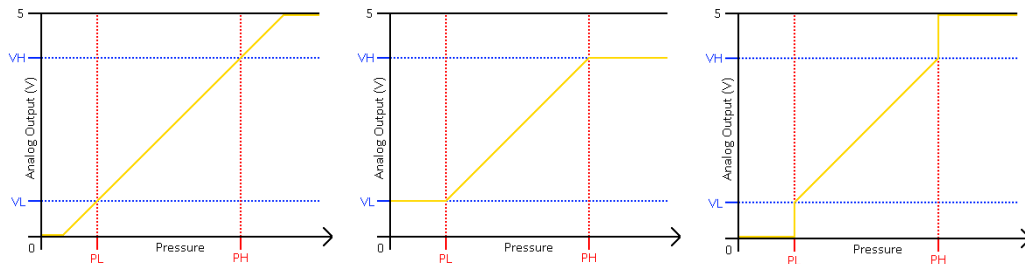
`#1234:TO:-18.14 //apply new temperature offset (C)`
`$1234:TO:C:-18.16`

`#1234:TO //confirm new temperature offset`
`$1234:TO:C:-18.14`

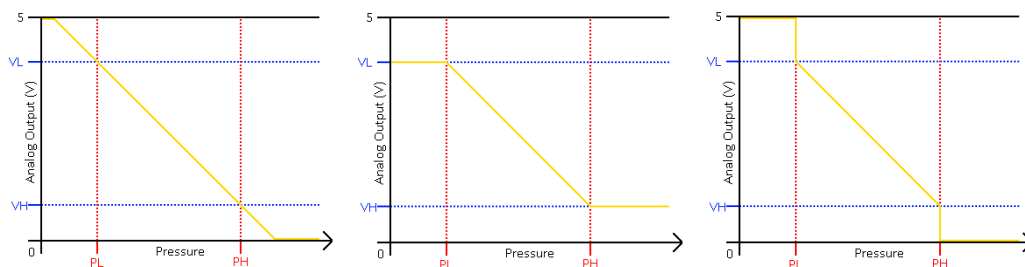
9. Analog Settings - AS

The AS command allows viewing and editing of the transmitters analog output settings. The transmitter uses 4 set points for pressure and voltage to adjust the analog output. These set points can be adjusted by modifying 4 values, pressure low (PL), voltage low (VL), pressure high (PH) and voltage high (VH), which creates a gradient for the output to follow as pressure changes.

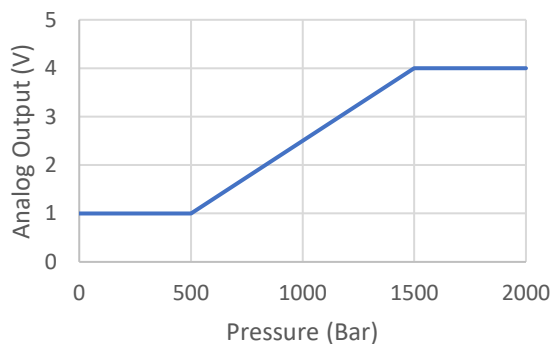
There is also a 5th setting which can be used to modify the head and tail of the output (should the output be set as anything other than 0-5V). To modify this setting, the operating mode modifier can be used (OM). These settings are linear mode (LM), voltage cap mode (VC) and pressure cap mode (PC) with matching graphs below respectively.



If voltage low is set higher than voltage high then the output will become inverted.



In the example below, the analog settings are viewed first, then set so that a 0-2000Bar transmitter has a 1V output from 0-500Bar, a 4V output from 1500-2000Bar and a 1-4V output between 500-1500Bar (such as below) and is then checked again.



Command: AS

Arg1: <None>, <Analog Setting Selection>

Arg2: <Analog Output Setting>, <Analog Output Mode>

Example: `#1234:AS //view current analog settings`
`$1234:AS:PL:Bar:0.0000`
`$1234:AS:PH:Bar:2000.0000`

```

$1234:AS:VL:V:0.0000
$1234:AS:VH:V:5.0000
$1234:AS:OM:LM

#1234:AS:PL:500 //set "pressure low" to 500Bar
$1234:AS:PL:Bar:500.0000

#1234:AS:VL:1 //set "voltage low" to 1V
$1234:AS:VL:V:1.0000

#1234:AS:PH:1500 //set "pressure high" to 1500Bar
$1234:AS:PH:Bar:1500.0000

#1234:AS:VH:4 //set "voltage high" to 4V
$1234:AS:VH:V:4.0000

#1234:AS:OM:VC //set more to "voltage cap"
$1234:AS:OM:VC

#1234:AS //confirm analog settings
$1234:AS:PL:Bar:500.0000
$1234:AS:PH:Bar:1500.0000
$1234:AS:VL:V:1.0000
$1234:AS:VH:V:4.0000
$1234:AS:OM:VC

```

10. Set Baud Rate - SB

The SB command allows the user to change the baud rate of the transmitter to a range of options. The baud rate can also be viewed to confirm correct settings. Can should be taken to adjust the interface settings to match the new baud rate setting of the transmitter.

If a baud rate that is not allowed is selected then the baud rate will not change and the current baud rate will be returned.

Command: SB

Args: <None>, 9600, 14400, 19200, 28800, 38400, 57600, 11520, 230400

```

Example: #1234:SB //confirm current baud rate
$1234:SB:57600

#1234:SB:9601 //applied incorrect baud rate
$1234:SB:57600 //baud rate not changed

#1234:SB:9600 //apply new baud rate
$1234:SB:9600

<Interface Settings Changed to 9600>

#1234:SB //confirm new baud rate
$1234:SB:9600

```

Units

This section covers the units that the transmitters are able to provide pressure and temperature readings in.

1. Pressure Units

The available pressure units are as follows (pressure units are case sensitive) with the argument data on the left and the description on the right:

- uBar - microbar
- mBar - millibar
- Bar - bar
- mPa - millipascals
- Pa - pascals
- kPa - kilopascals
- MPa - megapascals
- psi - pounds per square inch
- kpsi - 1000 pounds per square inch
- psft - pounds per square foot
- atm - standard atmosphere
- at - technical atmosphere
- Torr - Torr
- mTorr - millitorr
- cmAq - centimetres of water (4°C)
- mAq - meters of water (4°C)
- inAq - inches of water (60°C)
- ftAq - feet of water (60°C)
- mmHg - millimetres of mercury (0°C)
- cmHg - centimetres of mercury (0°C)
- inHg - inches of mercury (0°C)
- kgscm - kilograms per square centimetre

2. Temperature Units

The available temperature units are as follows (temperature units are case sensitive) with the argument data on the left and the description on the right:

- C - celsius
- F - fahrenheit
- K - kelvin

Useful Tools

Below are some useful tools for debugging and communicating with an ESI RS-485 transmitter using an RS-485 adapter or PC port.

1. FTDI Chip “USB-RS485-WE-1800-BT” USB to RS-485 Adaptor Cable

The cable allows direct communication between a host PC and RS-485 devices if connected to the bus correctly through either a custom or standard D-sub connector. Note that only the ground, RS-485 A and RS-485 B should be connected. The positive pin supplies +5V and should not be connected to the transmitter power supply.

2. RealTerm: Serial Capture Program

RealTerm is a useful piece of software which allows serial communication through a PC's COM ports if using a PCle or USB adaptor such as above. RealTerm gives the option for many baud rates, ASCII readouts and logging as well as a port echoing feature which allows interception of serial data.

3. ESI-USB Software

The ESI-USB software allows connection of ESI RS-485 transmitter (with a valid adaptor) as well as the ESI USB based transmitters. With datalogging capabilities, graphing and multiple channels, the software offers an interface for easy pressure monitoring.

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This document was created to support the use of the following RS-485 Transmitter equipment:

- GS4400B____AB
- GS4400B____AC
- GS4400B____AM
- GS4400B____AN

Operation of this equipment should always be carried out by trained personnel and in accordance with the manufacturers' guidelines. Failure to do so will be at the users own risk. ESI Technology will not be held liable for any losses and/ or damages incurred from improper use of equipment.

ESI Technology Ltd operates a policy of continuous product development. We reserve the right to change specification without prior notice. All products manufactured by ESI Technology Ltd are calibrated using precision calibration equipment, traceable to national measurement standards.

****Information provided within this manual is current and correct as of 17/06/2021. Please refer to the ESI website (esi-tec.com) or contact the office (info@esi-tec.com) for any updates****